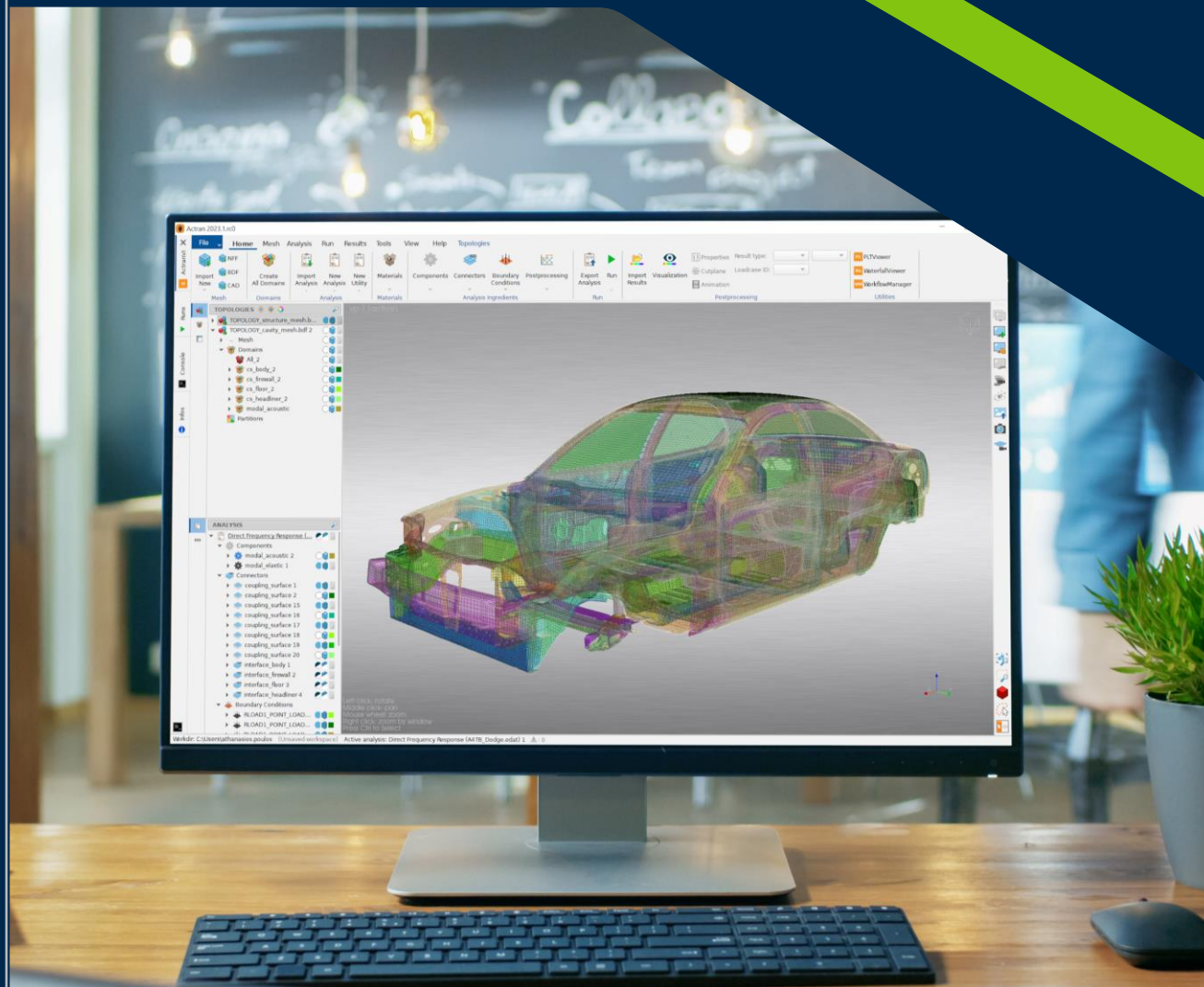




Actran

Release 2025.2:

What's new



Introduction to Actran

Actran | The most complete Acoustics CAE suite on the market

25+ years of development

- Complete solution for acoustic, vibro-acoustic and aero-acoustic simulation with dedicated pre and post processing environment including vertical solutions.
- Aiming at providing the most technologically advanced acoustic simulation capabilities to our customers. Covering different frequency ranges, physics and components (e.g., acoustic treatments).
- Seamless connections with third party software for tackling multi-physics problems.
- Efficient implementation and highly scalable solver running on local machines, HPC and cloud (CPU & GPU).

**We empower every
engineer with ground-
breaking acoustic simulation
to design a quieter and more
sustainable world**



Release summary

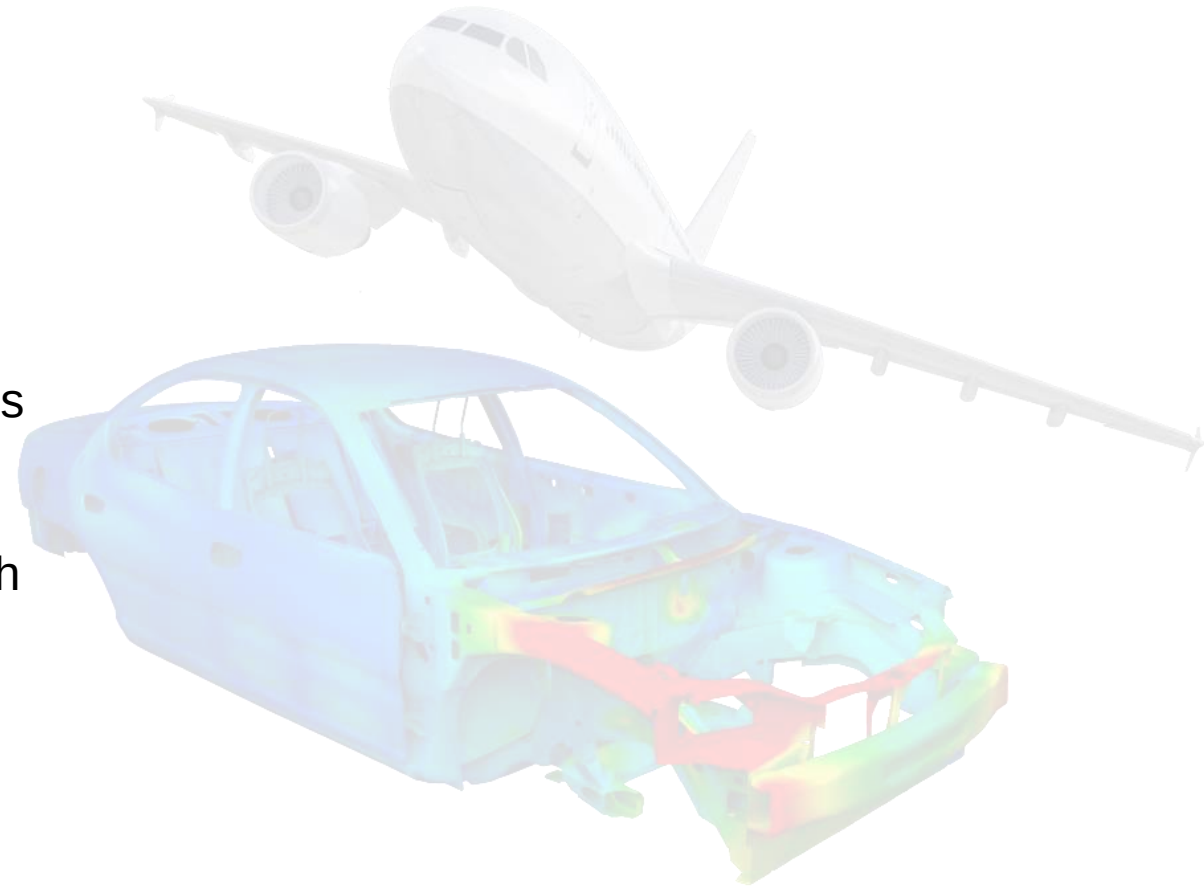
Actran release 2025.2

Release highlights

Major updates and enhancements in Actran 2025.2

Header

- Tackle large space and high-frequency noise challenges using the new integrated radiosity solver.
- Spend less time modelling ePowertrain noise with this simple, streamlined workflow manager. Quickly compute the effectiveness of acoustic cover treatments or use equivalent sources for analysis with the enhanced workflow.





Radiosity

Acoustic radiosity solver in Actran

Description:

The **radiosity solver**, a geometric-energetic-based acoustic approach, tackles **large space and high-frequency noise** propagation.

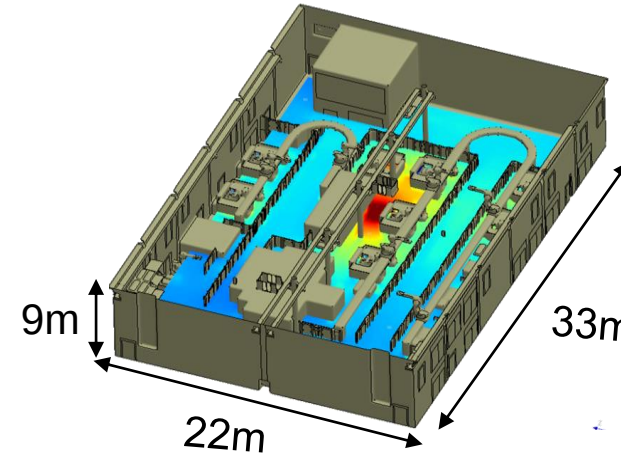
The radiosity method is ideal for analysing acoustic energy in large domains, which are challenging for wave-based approaches. Applications include vehicle cabin, aircraft cabins, factory noise, noise propagation in trains, and exterior radiation.

Benefits:

- **Ease of use:** Radiosity is a surface-based approach which simplifies model building.
- **Efficiency:** Effectively address high-frequency and large-domain acoustic problems with minimal computational effort.
- **Performance:** Use a rendering GPU to enhance computational performance with less dependence on the frequency of the solution.

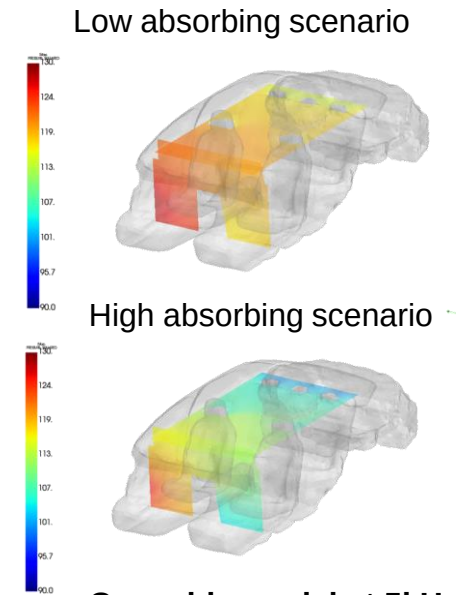
Who needs it:

- NVH engineers handling high frequency and large domains



Factory noise at 5 kHz

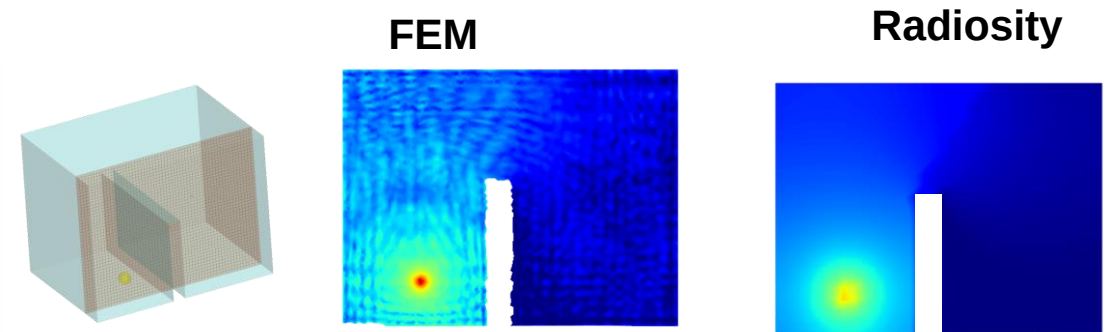
Performance ~ 10 min



Car cabin model at 5kHz

Performance ~ 1 min

Validation case study



FEM vs radiosity comparison shows strong correlation in 4.4–5.6 kHz range

Radiosity workflow in ActranVI

Description:

ActranVI supports full pre-processing for radiosity model setup, streamlining the workflow for radiosity analysis.

Key capabilities include:

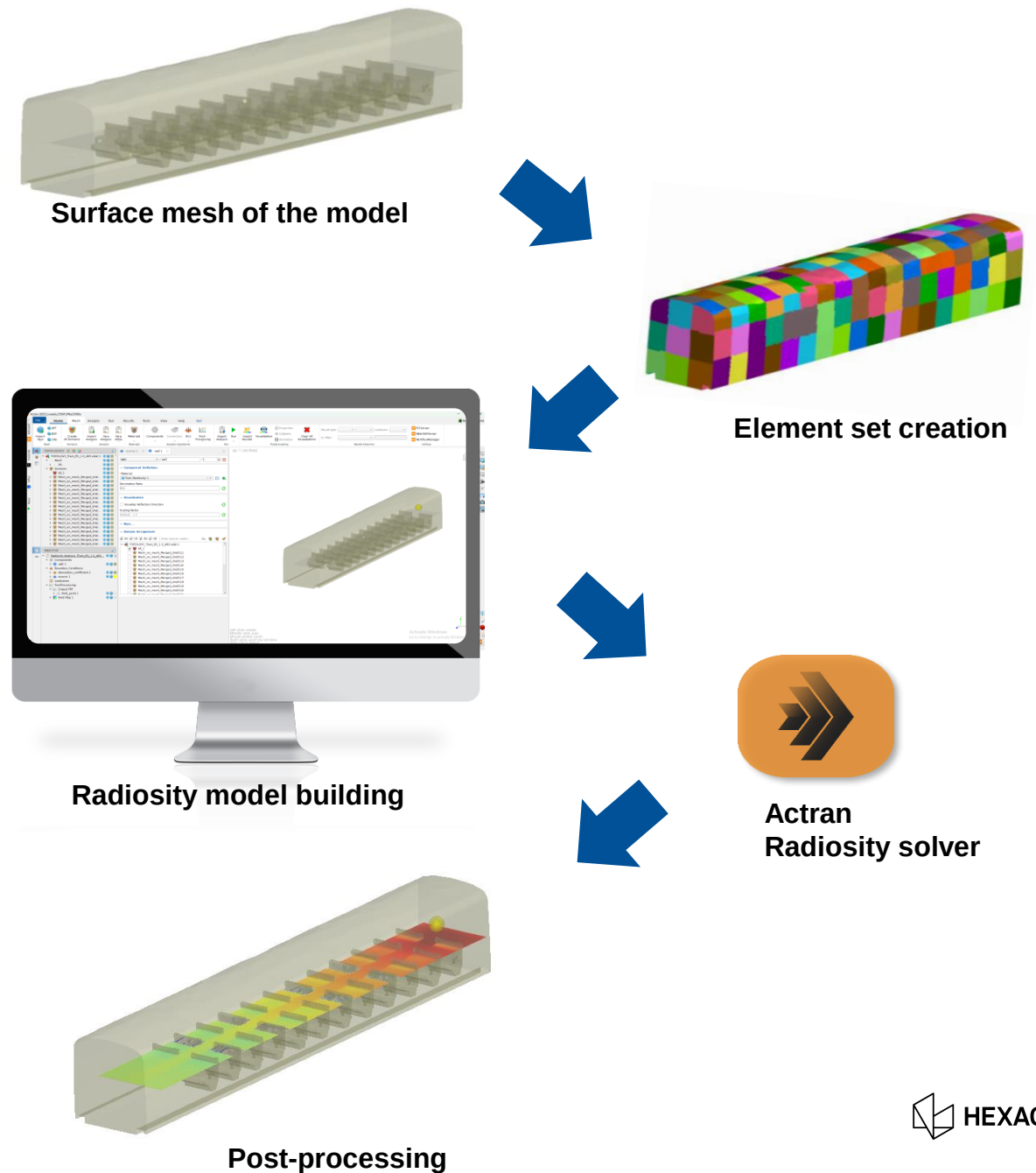
- Surface mesh generation
- Element set creation
- Normal orientation tools
- A dedicated element set creation tool for simplified radiosity model building

Benefits:

- **Ease of use:** Build and manage radiosity models directly within ActranVI, such as the creation or modification of boundary conditions, excitation, etc.
- **Simplified interface:** Use a single interface to pre-process and post-process the data.

Who needs it:

- NVH engineers handling high frequency and large domains





ActranVI, PLTViewer and meshing

Smooth, scalable UI for every screen

Description:

The new interface supports modern resolutions (1080p to 4K) and fractional scaling (125%, 150%, 175%).

UI elements scale smoothly across all Actran apps, including the VI, PLT viewer, the waterfall viewer, and the workflow manager.

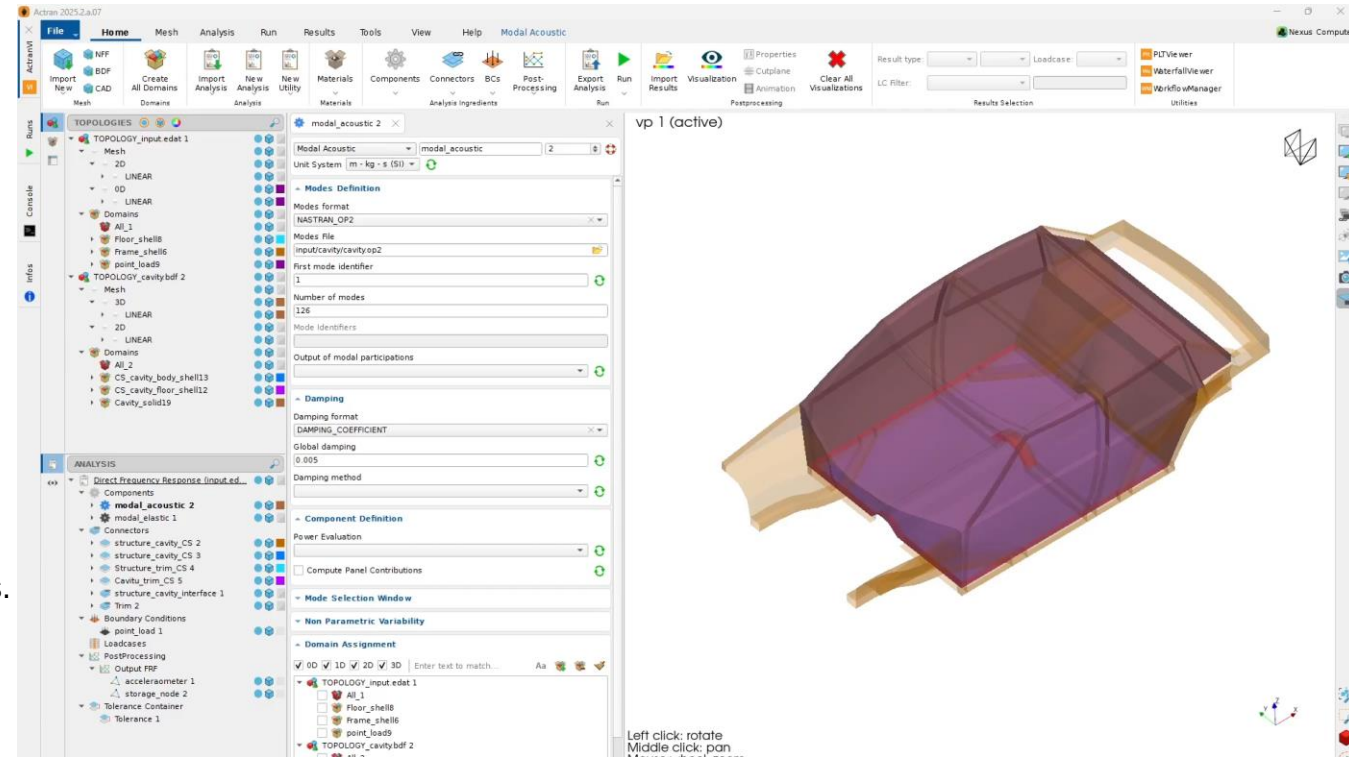
The interface enables seamless behaviour when moving apps between screens with different scaling.

Benefits:

- **Intuitive:** Get a sharp, readable UI on all modern displays.
- **Familiar tools:** Enjoy a consistent user experience across devices.
- **Productivity:** Use high-DPI and multi-monitor setups.

Who needs it:

- All Actran users



Mesh editor framework

Description:

The mesh editor is a **new framework dedicated to meshing operations**. Actran 2025.2 includes the first version of the mesh editor. The mesh editor allows users to:

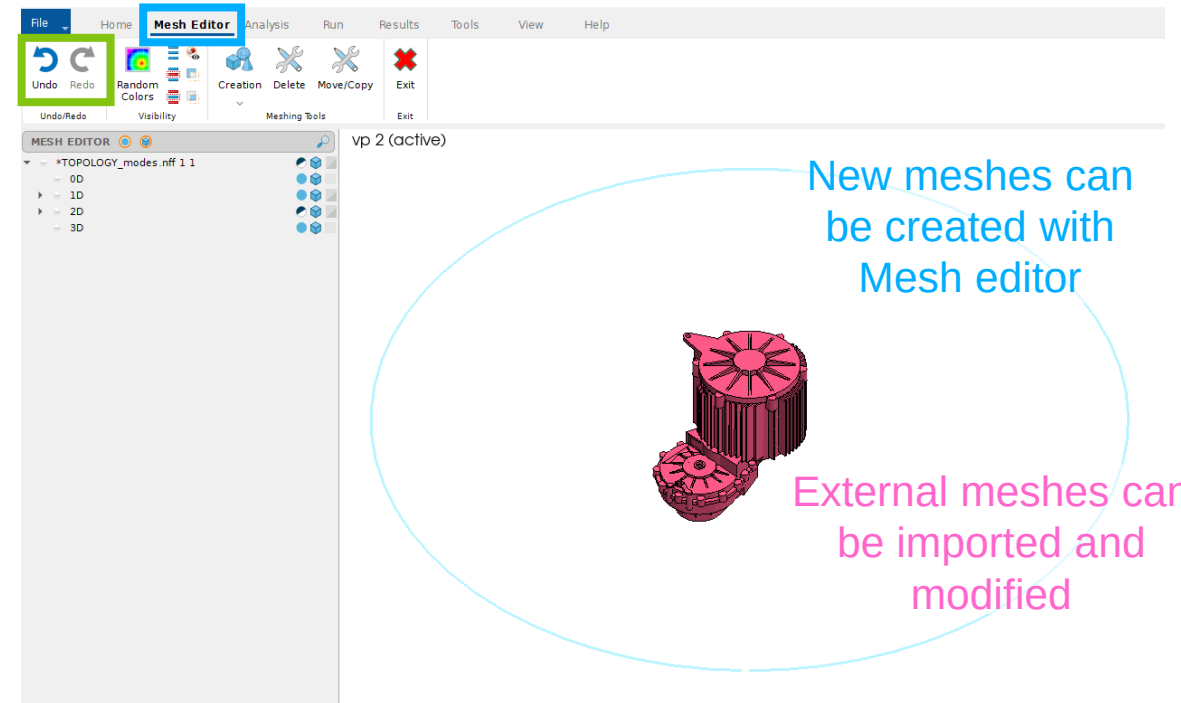
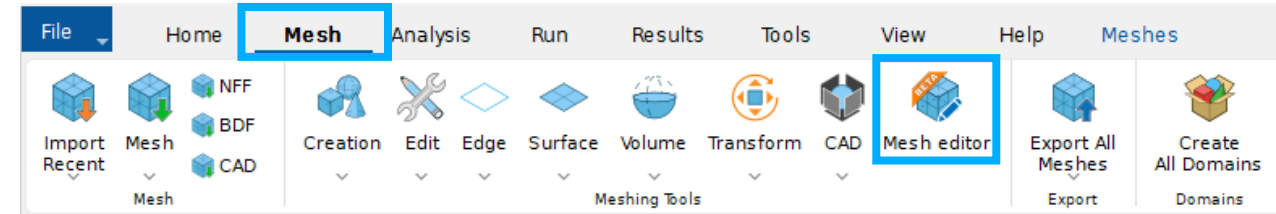
- Import existing meshes
- Create new meshes
- Modify meshes:
 - Edit PIDs: move, copy, or delete

Benefits:

- **Efficiency:** Create meshes faster and more easily.
- **Intuitive interface:** The interface is the same as traditional meshing tools.
- **Control:** Use undo/redo controls during mesh edition for better control of meshes.

Who needs it:

- All users performing meshing operations with Actran



Acoustic indicators tool: oblique incidence and DSF excitation

Description:

The acoustic indicator tool predicts the sound attenuation capabilities of multi-layered materials with a Transfer Matrix Method (TMM) approach.

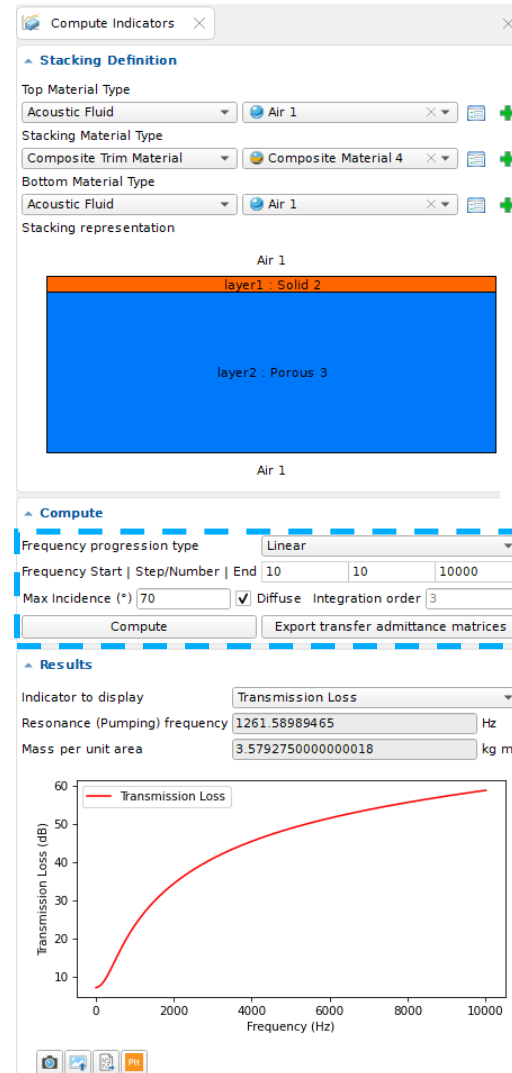
Transmission Loss (TL) and the absorption coefficient (α) of multi-layered materials can be assessed in the tool, now from an incident wave with an angle of incidence or a Diffuse Sound Field (DSF).

Benefits:

- **Fast:** Get results in under one second.
- **Simple interface:** Use a straightforward workflow in a single window.
- **Rapid iteration:** Use scripts to change one text variable instead of starting from scratch to make quick adjustments for sensitivity analysis.

Who needs it:

- Any user who wants to quickly assess a complex multi-layered (or single layer) material acoustic characteristics.

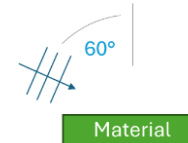
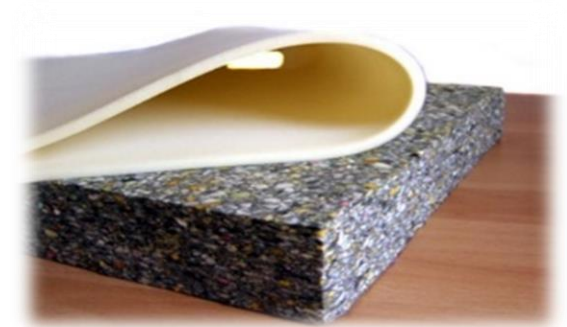


Materials

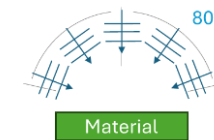
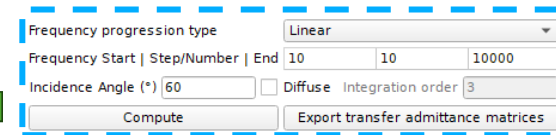
Visualization

Parameters

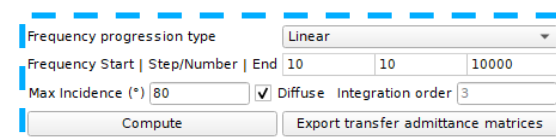
Results



Oblique incidence



DSF excitation





NVH

Support of analytical trims in the new Nastran PEM workflow

Description:

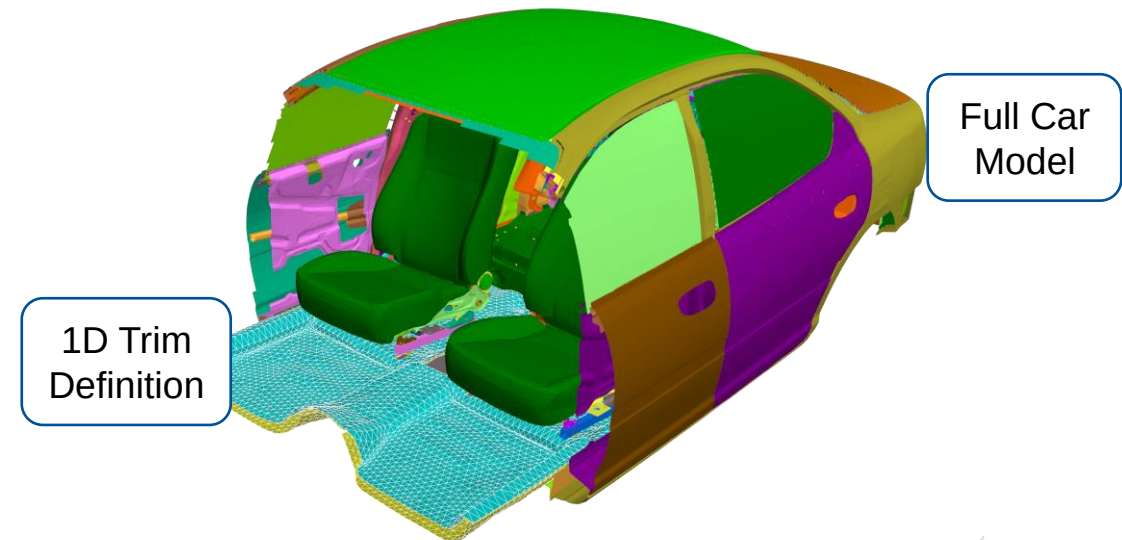
A new enhanced Nastran PEM workflow was introduced in the Actran and Nastran 2024.1 release. However, this functionality was previously limited to 3D meshed trims. With Actran and Nastran 2025.2, analytical (1D) trims are also supported.

Benefits:

- **Faster meshing:** Analytical trims don't require a complete 3D mesh, just top and bottom surfaces.
- **Ease of use:** Set up the trim model much more easily and quickly.
- **Easy iteration:** Change trim layers without any remeshing.

Who needs it:

- All Nastran PEM users.



Local damping support for space applications

Description:

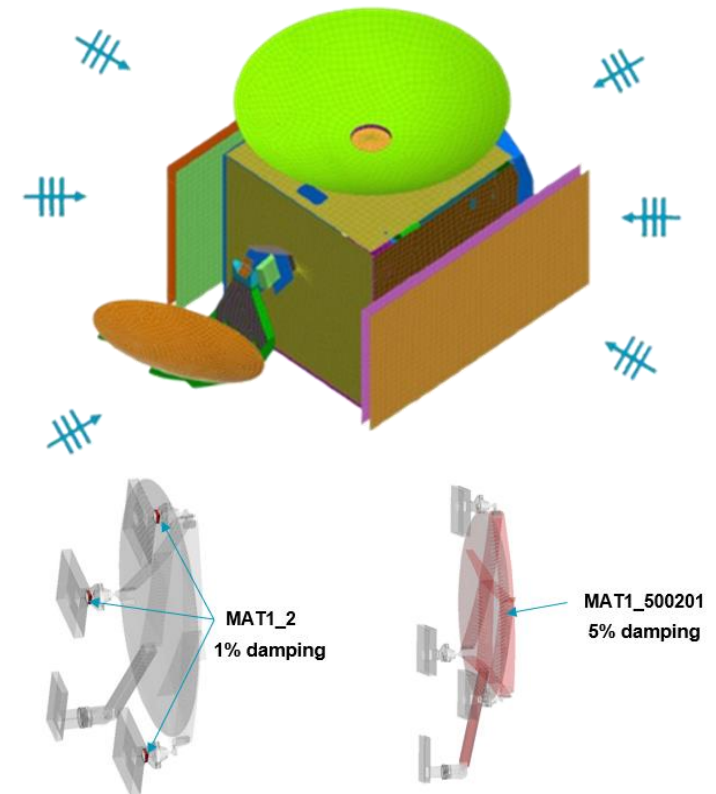
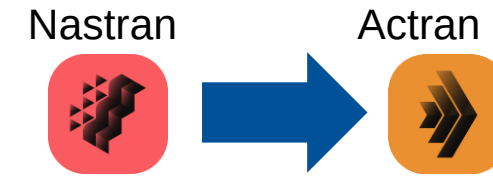
Define local damping coefficients for different PIDs or materials for coupled vibro-acoustic simulation using the Space workflow manager.

Benefits:

- **Greater realism:** Get local damping results for more realistic results
- **MSC Nastran interoperability:** It is easy to implement for MSC Nastran users, and can be used with both SOL103 and SOL111 input files.
- **Intuitive to use:** No new syntax is used for this capability, making it familiar and intuitive to use.

Who needs it:

- Users of the space workflow manager



Seamless non-structural mass handling

Description:

A new septum component allows users to easily include non-structural mass and surface stiffness in the analysis.

This component also ensures smooth conversion of MSC Nastran's NSM cards into Actran models.

Benefits:

- **MSC Nastran integration:** This component simplifies the translation of Nastran NSM cards into Actran and allows users to directly import MSC Nastran models with septum components.
- **Consistency:** Enjoy improved consistency between Actran and Nastran analysis results.

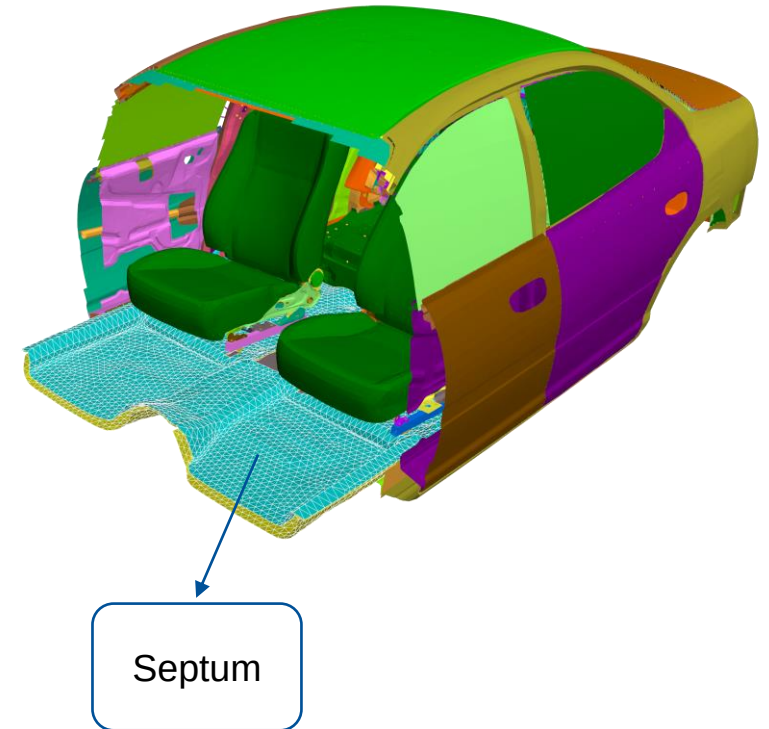
Who needs it:

- Automotive OEMs managing trim mass layers
- Aerospace users who require accurate mass modelling on panels and surfaces
- Any engineer working with MSC Nastran-to-Actran workflows who wants seamless conversion and matching results

Nastran



Actran



SIF indicator to assess the acoustic diffusivity

Description:

An affordable and accessible solution to produce a diffuse acoustic field is the Direct Field Acoustic Testing (DFAT), which relies on loudspeaker columns placed around a satellite system.

The Sinc Indicator Function (SIF) output, between 0 and 1, assesses the quality of the acoustic diffusivity of a field.

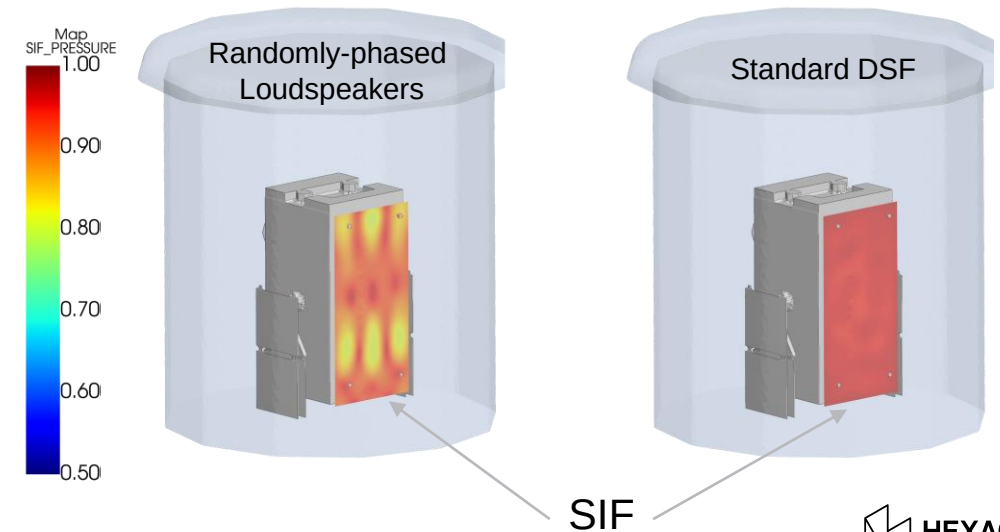
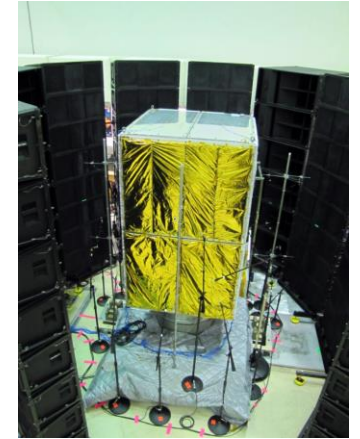
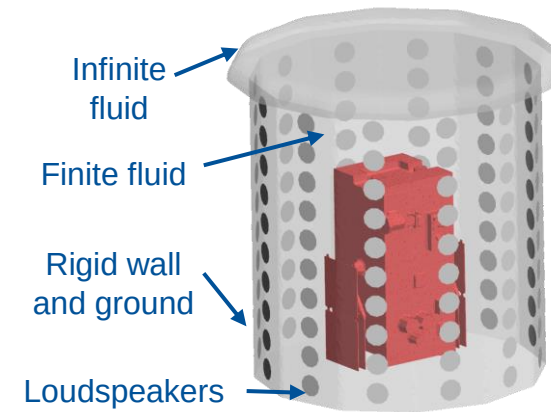
The SIF compares the spatial cross-correlation of this field to a perfect diffuse sound field.

Benefits:

- **Understand the effects of changes:** Evaluate the impact of any change in the setup on the diffusivity of the created acoustic sound field.
- **Clear KPIs:** Understand at a glance whether perfect diffuse sound fields have been achieved.

Who needs it:

- DFAT manufacturers and users.
- All users working in the space industry





Aeroacoustics and turbomachinery

Ansys Fluent .h5 files for aero-acoustic computations

Description:

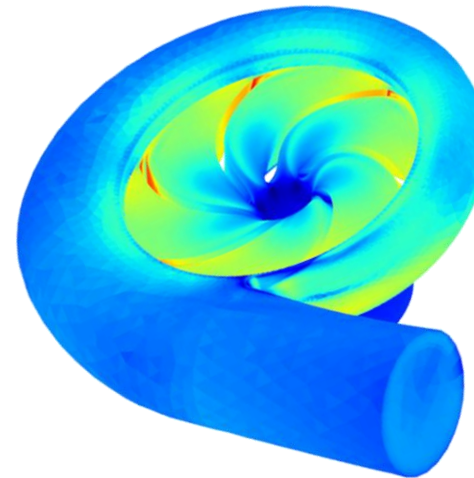
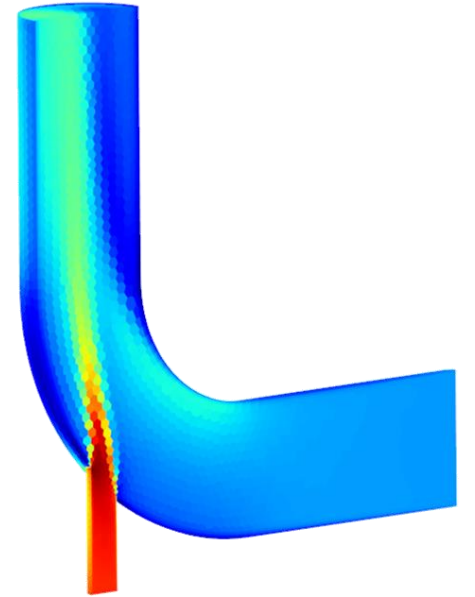
The ICFD utility for reading imported CFD data in Actran now supports .h5 file formats exported by default from Ansys Fluent. Actran can use Ansys Fluent .h5 files for aero-acoustic source and mean flow computations.

Benefits:

- **Interoperability with Ansys Fluent:** Support for .h5 files simplifies data exchange with Ansys Fluent for Actran users.

Who needs it:

- All aero-acoustics users



Automated mesh creation adapted to mean flow

Description:

The flow remeshing feature allows users to automatically generate a mesh according to the spatial heterogeneous flow that fulfils the convected acoustic meshing criteria. To feed this feature, the user must provide the CFD mean flow and closed acoustic domain information.

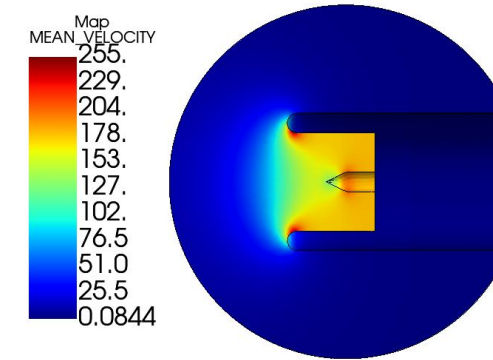
Benefits:

- **No CAD preparation required:** There is no longer a need for tedious CAD model preparation to identify equivalent speed areas.
- **Faster results:** The model-building time for a nacelle inlet is reduced from days to hours.
- **Automate remeshing:** Use the ICFD analysis to automate acoustic remeshing when frequencies change and automatically create a projected CFD mean flow database.
- **Accurate meshes:** Use additional options to improve mesh accuracy, such as local mesh refinement and curvature surface rebuilding.

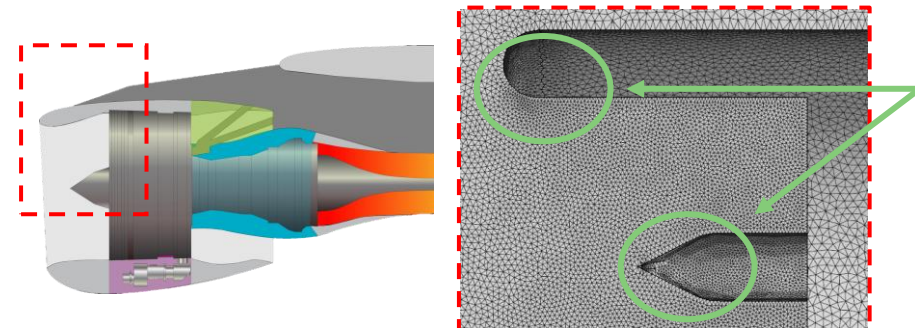
Who needs it:

- Actran TM and aeroacoustics users dealing with heterogeneous CFD mean flow in complex models such as aircraft nacelles

CFD Mean flow velocity



Adaptative acoustic mesh



Refined mesh areas due to high mean velocity.

Water material in Actran DGM

Description:

Water is now available as a material in Actran DGM. This new feature allows users to handle water-specific properties adapted to underwater simulations. Use temperature and salinity fields to compute acoustic water properties (fluid density and speed of sound).

Benefits:

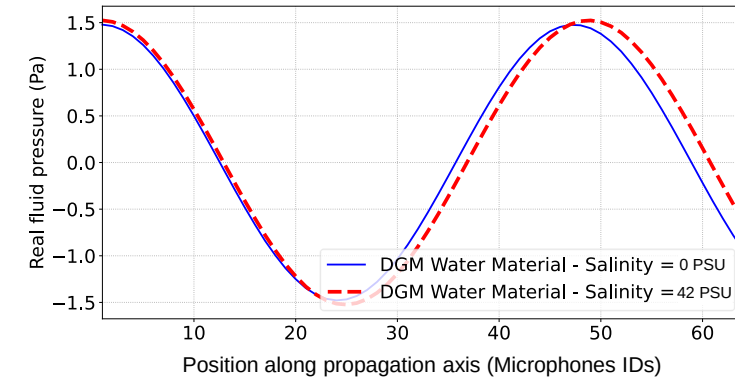
- **Accuracy:** Compute Actran DGM underwater simulations with great accuracy.
- **Engineering speed:** Save engineering time with quick and easy material-based computations



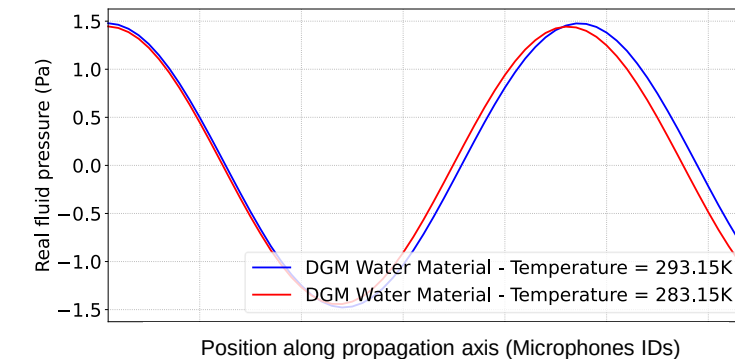
Who needs it:

- Actran DGM hydrodynamics users dealing with submarine, boat and propeller models immersed in water

Salinity impact on pressure



Temperature impact on pressure





Workflow manager

UX improvements in the ePowertrain noise workflow

Description:

Improvements to the existing ePowertrain noise workflow (root cause analysis) include:

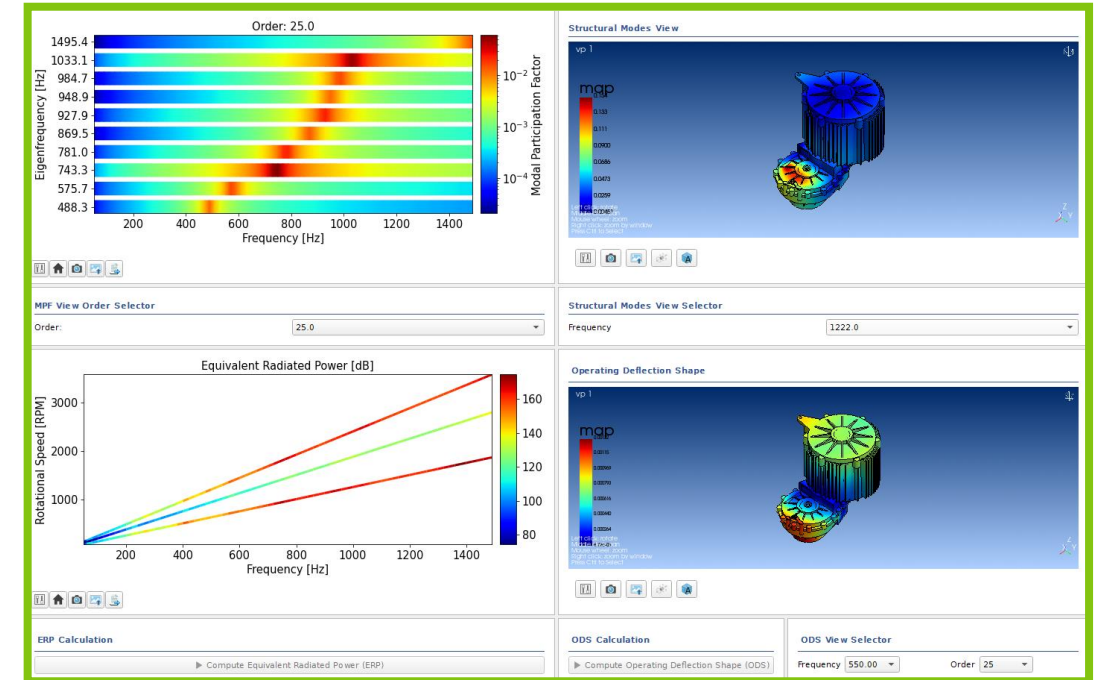
- The ability to handle modal participation factors based on displacement or velocity
- Automatic detection of intersections in meshes (created meshes can also be exported to be modified with another tool)
- Global results for element contribution

Benefits:

- **Flexibility:** More inputs are supported, and mesh creation is more easily controlled.
- **Improved confidence:** Additional results for root cause analysis give more clarity to the results.

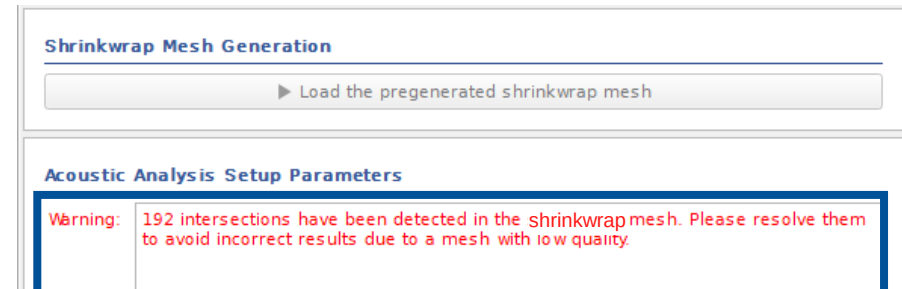
Who needs it:

- Powertrain NVH engineers
- Romax Spectrum Acoustic users



All figures are visible at once in the structural analysis tab

Automatic
detection of
intersections



Equivalent sources in the ePowertrain noise workflow

Description:

The ePowertrain noise workflow can now assess equivalent sources. Computed equivalent sources can be used to represent the powertrain in later studies. For example, to perform a pass-by noise analysis. The computation of equivalent sources is optional for users only interested in the root cause analysis, and the existing workflow will not be impacted.

Benefits:

- **Ease of use:** There is no need to set up a specific acoustic analysis.
- **Efficiency:** Quickly evaluate equivalent source strengths for multiple sources and orders (the results of the already existing acoustic analysis are reused instead of running a new computation).
- **Export capabilities:** Use computed source strength in other analyses.

Who needs it:

- Powertrain NVH engineers
- Romax Spectrum Acoustic users

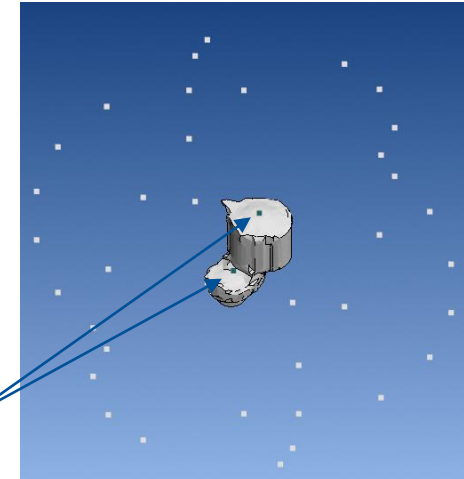
Acoustic Analysis Setup Parameters

- ▶ Analysis Parameters
- ▶ Fluid Properties
- ▶ Green Analysis Parameters
- ▶ Exterior Acoustic Parameters
- ▶ Tolerance Parameters
- ▶ Output Parameters: Microphone Positions

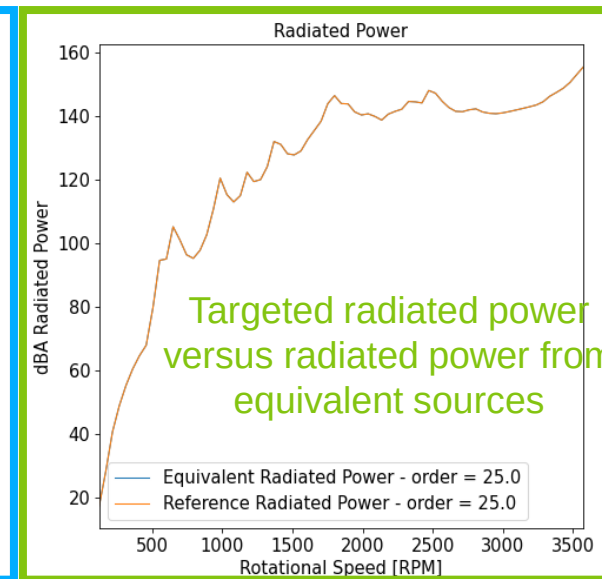
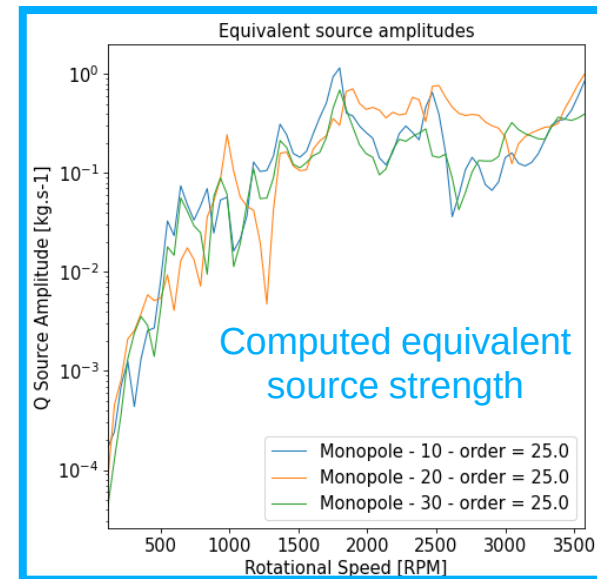
Equivalent Sources: Monopoles Positions

Monopole Source  From Source File 

Monopole Locations from File 



Structural Analysis Acoustic Analysis Root Cause Analysis **Equivalent Sources**



Acoustic cover treatments in the ePowertrain noise workflow

Description:

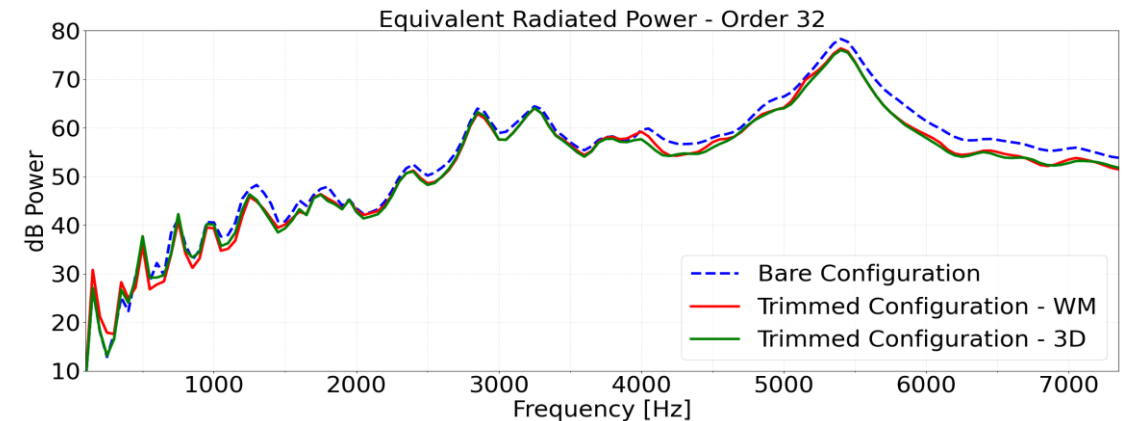
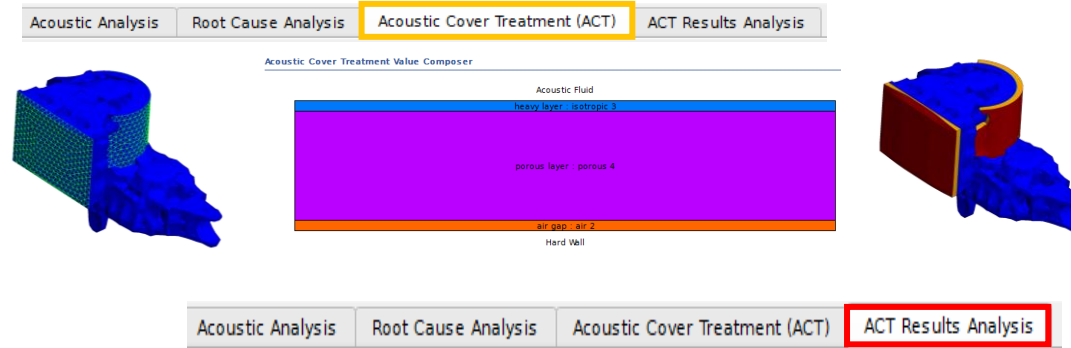
The ePowertrain noise workflow has been expanded with a new feature allowing users to quickly compute the insulation effects of the acoustic cover treatments, called encapsulations. This feature is optional for users only interested in the root cause analysis, and the existing workflow will not be impacted.

Benefits:

- **Ease of use:** There is no need to set up an acoustic analysis; just define the material properties for the cover(s) and select the region(s) where it should be located. There is no need to build the physical 3D trim.
- **Reduced RAM and time to results:** There is no need to run a new acoustic analysis for each treatment configuration.
- **Efficiency:** Quickly evaluate the effect of the acoustic cover treatments on global parameters (SWL) and local parameters (SPL), or check more configurations, and, if required, combine them together.

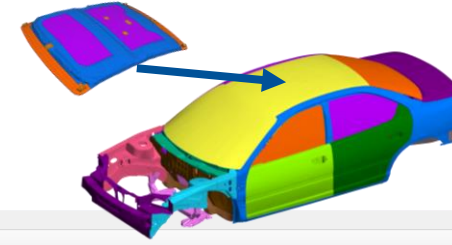
Who needs it:

- Powertrain NVH engineers
- Romax Spectrum Acoustic users



- 40% run time compared to a 3D method
- 60% RAM consumption compared to a 3D method

Acoustic trims in the SEA manager



Description:

Acoustic trims can now be easily computed and combined in SEA Manager.

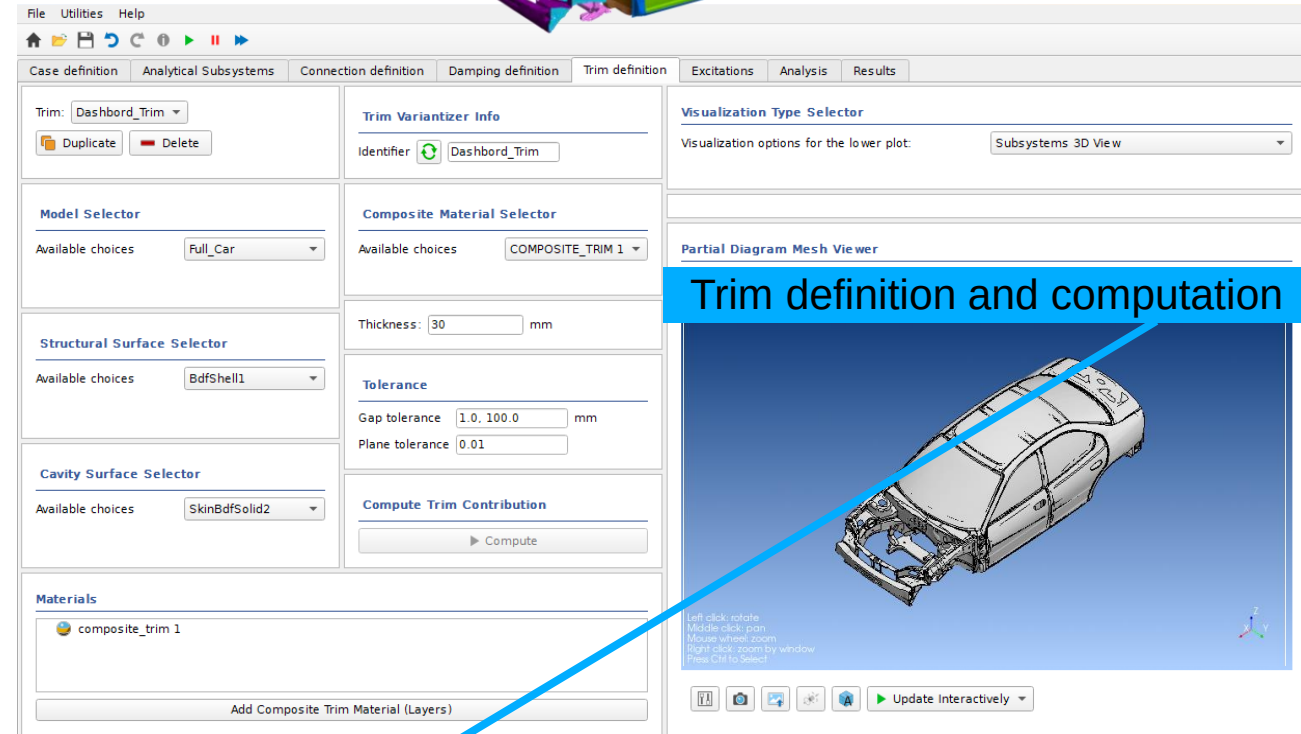
Once each trim effect is computed individually, creating new trim configurations that combine them takes seconds instead of hours.

Benefits:

- **Speed:** Create and interactively analyse different trim configurations very quickly.

Who needs it:

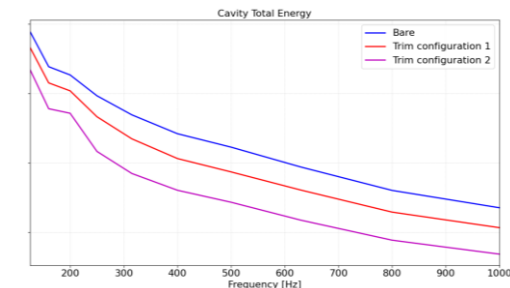
- Mechanical and acoustic engineers dealing with trims
- CAE engineers dealing with NVH mid- to high-frequency applications



Trim definition and computation

Easy creation of the trim configurations

Results analysis





Previous release

Actran Release 2025.1

Highlights from Actran 2025.1

ePowertrain Noise WorkflowManager:

A new ePowertrain workflow is available in WorkflowManager. Starting from structural modes and participation factors expressed in the order domain, the user will be able to build and run a complete acoustic radiation analysis in the order domain within a few clicks.

Bearing loads support and more improvements in the EM Noise WorkflowManager:

In addition to electromagnetic loads, bearing loads can now also be applied in the EM Noise workflow. Those bearing loads can be imported from CSV files exported by ROMAX or BDF files exported by Masta.

WorkflowManager contextual documentation:

Extensive documentation has been created for the most complex nodes of WorkflowManager. This documentation is accessible in an info icon ⓘ on the top right of the node.

Automatic SEA subsystems creation in Actran VI:

In Actran 2025.1, the grouping component of AUTO_SUBSYSTEMS sequence is now ported in ActranVI, to quickly view different subsystem configurations and make iterations much easier.

3D to 1D trim conversion tool: Increased robustness:

Since Actran 2024.1, existing 3D trims can be automatically converted to 1D trims. This conversion tool is accessible from ActranVI by right-clicking on a trim analysis and, selecting "Convert to Parametric 1D Trim". This conversion is now much more robust thanks to a new set of parameters, which allow better control of the 1D trim output.

ARM processor support:

Since Actran 2024.2, Actran supports ARM processor machines with a Linux operating system.



Additional resources

Actran release 2025.2

Additional Resources

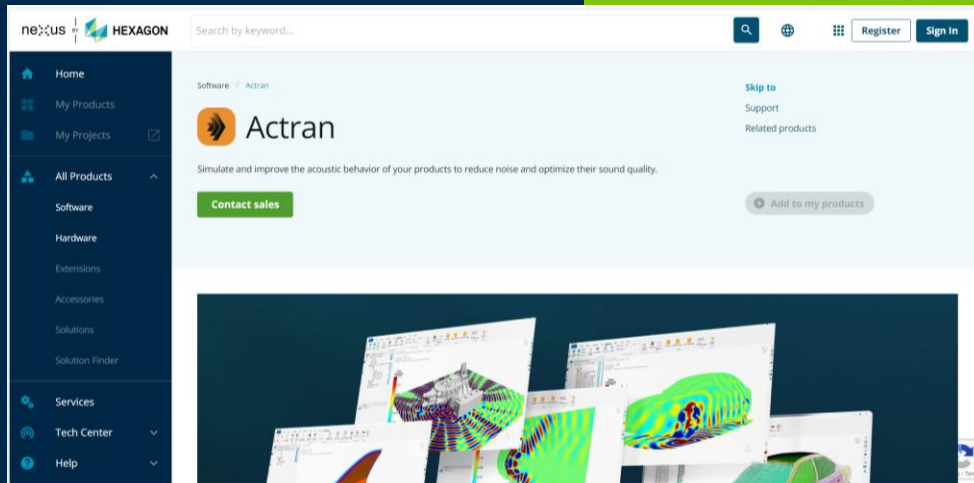


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